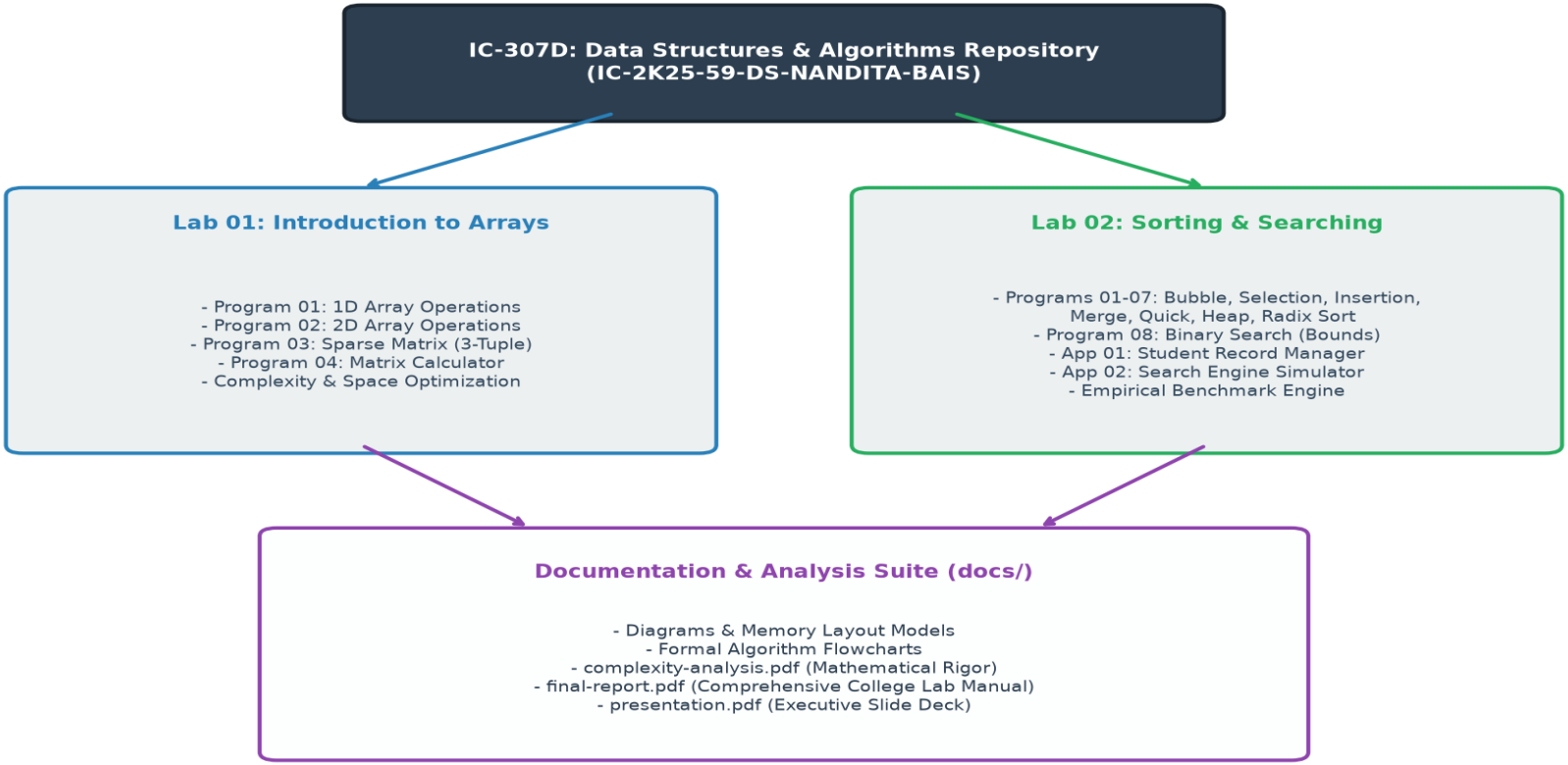
IC-307D: Data Structures & Algorithms Lab

Comprehensive Implementation of Lab 1 & Lab 2 Assignments

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Project Architecture & Repository Structure



Lab 01: Arrays & Sparse Matrix Optimization

Key Implementations in Unit 1:

- **Program 01 (1D Arrays):** Robust insertion, deletion, searching, in-place reversal, and rotation O(n).
- **Program 02 (2D Arrays):** Matrix row/col shifts, 90-degree rotations, spiral & boundary traversals.
- **Program 03 (Sparse Matrix):** Triplet 3-tuple format saving up to 75% memory on sparse data.

Implemented Fast Transpose reducing complexity from O(cols * non_zeros) to linear O(cols + non_zeros).

- **Program 04 (Matrix Calculator):** Dimension validation, multiplication O(n^3), determinant & inverse.

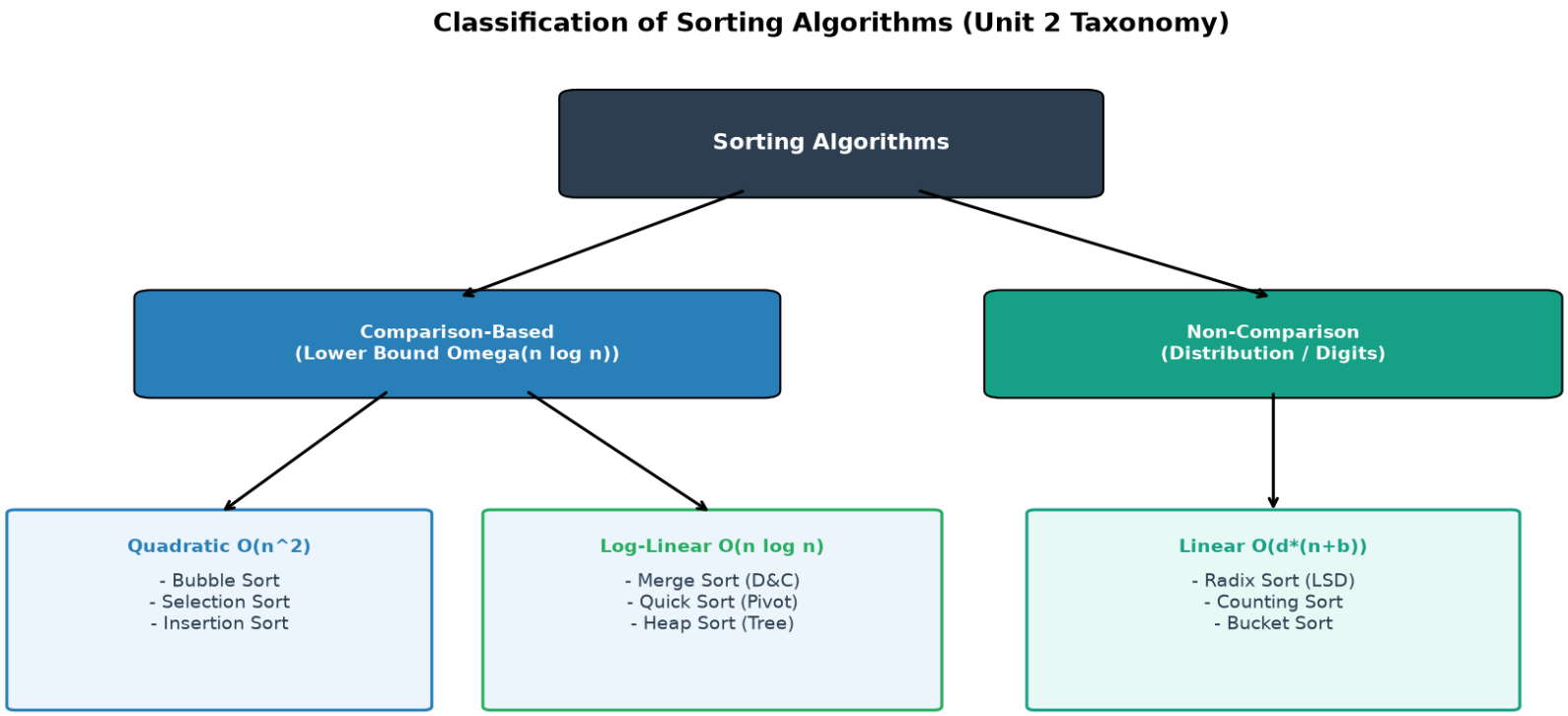
Standard 4x4 Matrix (16 Ints = 64 Bytes)
75% Zeros (Sparse)

0	0	5	0
0	8	0	0
0	0	0	3
7	0	0	0

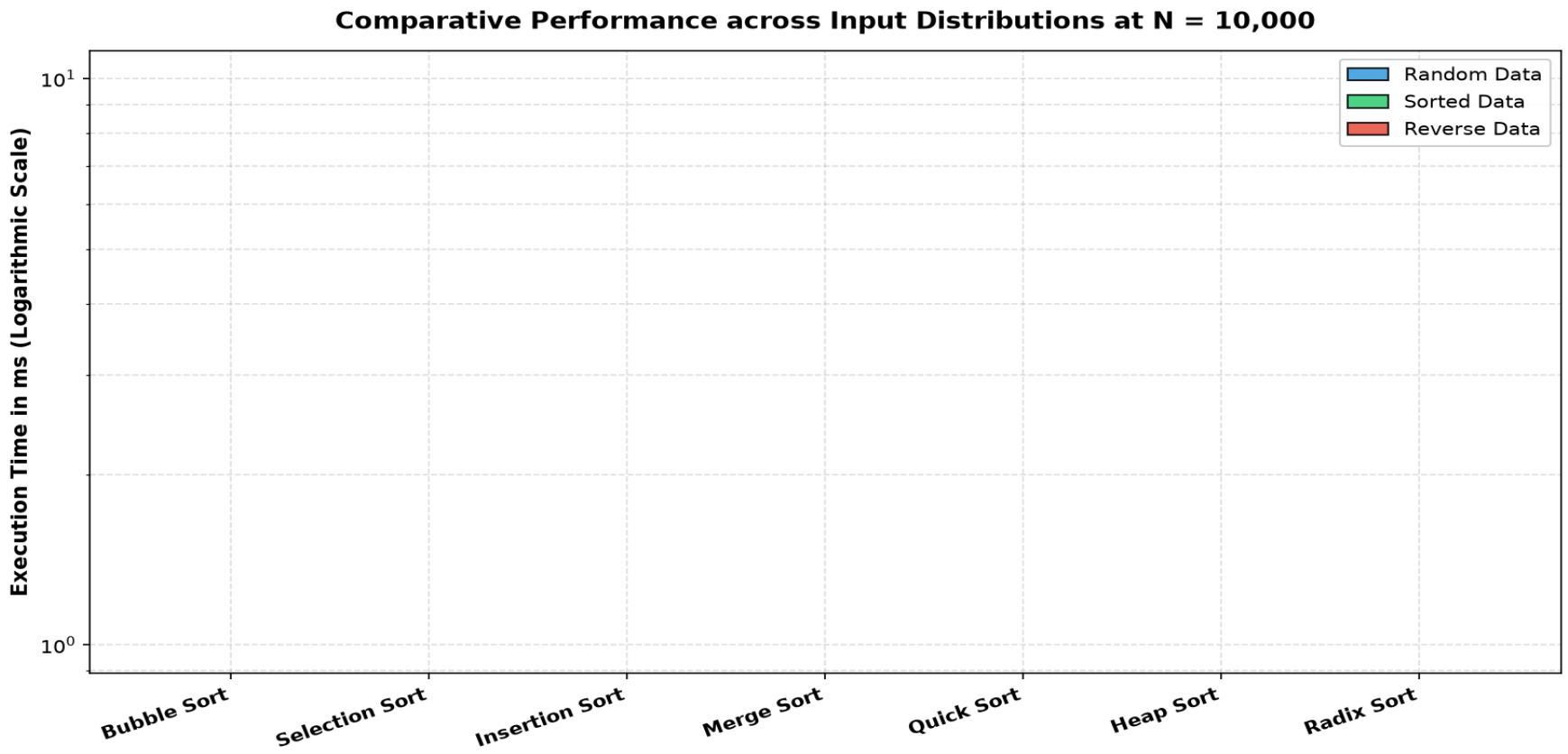
3-Tuple Representation (5 Rows x 3 = 15 Ints)
Saves Space on High Sparsity

Row	Col	Val
4	4	4 (Meta)
0	2	5
1	1	8
2	3	3
3	0	7

Lab 02: Sorting Algorithms Taxonomy & Implementation



Empirical Execution Benchmark (Random, Sorted, Reverse Data)



Conclusion & Key Learning Outcomes

Key Takeaways & Educational Achievements:

1. **Algorithmic Rigor:** Mastered formal time & space complexity proofs across arrays, recursion, and sorting.
2. **Real-World Systems:** Engineered practical applications (Student Record Manager, Search Engine Simulator).
3. **Empirical Validation:** Proved theoretical big-O bounds through high-resolution benchmark timing and visualizations.
4. **Professional Tooling:** Built clean modular C codebase adhering to GCC standards, VS Code task automations, and comprehensive documentation.

Thank You! Questions & Discussions Welcome.